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AI PRINCIPLES AND PRACTICE · WEEK 8

Week 8 Summary

The mathematics that matters: everything to keep, in plain English

IN PLAIN ENGLISH

How to use this summary. The ten points below follow the lecture in order. Read them (about 20 minutes), try the ten questions without looking (about 15 minutes), then mark them. Re-read only the points you got wrong. Your capstone's PCA via SVD, its loss and its audit all build on these.

KEY IDEA

If you remember one thing. Three questions explain why your methods work. Shape: the SVD finds the directions that matter. Surprise: your loss is unavoidable surprise plus avoidable surprise, and training shrinks the avoidable part. Space: real data lives on a thin, low-dimensional sheet.

Part 1 The ten points

1.1 A matrix moves points; eigenvectors only get longer

A matrix is a table of numbers; multiplying an arrow by it stretches and usually turns it. A few special directions are only stretched: the **eigenvectors**, with their stretch factors, the **eigenvalues**: $A\,v = \lambda\,v$. For A = [[3, 1], [1, 3]], (1, 1) becomes (4, 4), stretched ×4, and (1, −1) becomes (2, −2), stretched ×2; (1, 0) becomes (3, 1), which has turned.

1.2 The SVD: every matrix is turn, stretch, turn

Any table of numbers, of any shape, can be written $A = U\,\Sigma\,V^T$: a turn (V^T), a stretch (Σ) and another turn (U). The stretches, $\sigma_1 \geq \sigma_2 \geq \ldots$, are the **singular values**, listed biggest first. The SVD lists the directions a table stretches most.

1.3 PCA is the SVD of your centred data

Subtract each column's average (**centre**), take the SVD: the rows of V^T are the directions, and the spread along direction k is $\sigma_k^2 \div (n - 1)$. For 300 people's height and arm span, direction 1 is about (0.69, 0.72), "overall size", with variance 177.6 against 5.7: 97% of the spread. You write this in five lines in the lab, and reuse it in your capstone.

1.4 Low rank: keep the big pieces

The SVD splits a table into pieces, biggest first. Keeping k of them is a **low-rank approximation**, and no other k pieces get closer (Eckart–Young). Each piece of a 120 × 180 photo costs 120 + 180 + 1 = 301 numbers, so k = 20 keeps 28% of the numbers with an error of 4.9%. **LoRA** (Week 9) uses the same idea: $W' = W + BA$, so a rank-8 change to a 768 × 768 layer learns 12,288 numbers instead of 589,824 (about 2%).

1.5 Surprise and entropy

The surprise of an event is $-\log_2 p$: 1 bit for a coin, 6 bits for a 1-in-64 event, 0 for a certain one. **Entropy**, $H = -\Sigma\, p \log_2 p$, is the average surprise: 1 bit for a fair coin, 0.47 bits for a 90/10 coin. Log base 2 gives bits; the natural log, used by PyTorch, gives nats (1 nat ≈ 1.44 bits).

1.6 Cross-entropy and KL: the cost of wrong odds

Cross-entropy, $H(p, q) = -\Sigma\, p \log_2 q$, is the average surprise when the real odds are p but you believe q. **KL divergence** is the extra part: KL = H(p, q) − H(p). Real weather 50/30/20, a forecaster believing 70/20/10: entropy 1.49 bits, cross-entropy 1.62, KL 0.13. KL is never negative, is zero only when q = p, and is not symmetric.

1.7 Your loss is the average surprise

The classification loss you trained with is cross-entropy: the average surprise at the right answer, in nats. Week 7′s moons network went from 0.77 to 0.07; a coin toss scores ln 2 = 0.69. The loss is unavoidable surprise (entropy) plus avoidable surprise (KL), and training can only shrink the second, so it pulls the model's odds towards the real ones; this is also **maximum likelihood**. On the credit data, logistic regression pays 0.472 nats against 0.528 for "22% default": the 23 features tell it about 0.08 bits per customer.

1.8 Mutual information and the bottleneck

Mutual information is how much one thing tells you about another: I = H(before) − H(after). Rain on 30% of days; dark clouds on 90% of rainy and 20% of dry days: 0.881 − 0.551 = 0.33 bits. The **information bottleneck** is a useful picture of hidden layers: keep what predicts the answer, forget the rest.

1.9 High dimensions are strange

In d dimensions, the outer 10% “peel” of an orange is $1 - 0.9^d$ of it: 19% in 2-D, 65% in 10-D, 99.997% in 100-D. Distances squeeze together: for 400 random points, the farthest pair is 593 times the nearest in 2-D but only 1.19 times in 1,000-D. This **curse of dimensionality** hurts nearest neighbours and K-means.

1.10 Real data lives on a thin sheet

The **manifold hypothesis**: real data lies near a thin, low-dimensional surface, like a rolled-up sheet of paper in a room. A rough **intrinsic dimension** is the number of PCA directions needed to keep 90% of the spread: handwritten digits need 21 of their 64 numbers, random noise 56 of 64, the credit data 13 of 23. A hidden layer flattens the sheet until a straight line can cut it.

Part 2 Ten questions

1. For A = [[2, 0], [0, 5]], which arrow is an eigenvector with eigenvalue 5?

A. (1, 0)
B. (0, 1)
C. (1, 1)
D. (5, 2)

2. What does the SVD say every matrix does?

A. rotate only
B. stretch only
C. turn, stretch, turn
D. add, multiply, add

3. Why do you centre the data before taking its SVD for PCA?

A. to make it faster
B. so the first direction follows the spread, not the average
C. so all variances become 1
D. to remove outliers

4. A 100 × 200 picture, keeping k = 10 pieces. What share of the numbers is stored?

A. 5%
B. about 15%
C. 50%
D. 10%

5. What is the surprise of an event with probability 1/8?

A. 0.125 bits
B. 1 bit
C. 3 bits
D. 8 bits

6. Which has the highest entropy?

A. a coin that lands heads 99% of the time
B. a fair coin
C. a coin that lands heads 90% of the time
D. a two-headed coin

7. Cross-entropy is 2.0 bits and entropy is 1.5 bits. What is the KL divergence?

A. 3.5 bits
B. 0.5 bits
C. −0.5 bits
D. 1.33 bits

8. A two-class model's loss is 0.69 nats. What does that mean?

A. it is perfect
B. it is no better than a coin toss
C. it is 69% accurate
D. it has overfitted

9. In 100 dimensions, what share of an orange is in its outer 10%?

A. 10%
B. about 50%
C. about 90%
D. over 99%

10. Digits need 21 of 64 directions to keep 90% of their spread; noise needs 56. What does this show?

A. digits are noisier
B. the digits live on a low-dimensional sheet
C. PCA fails on noise
D. noise has fewer dimensions

Part 3 Answers

1. **B.** $A \times (0, 1) = (0, 5) = 5 \times (0, 1)$. (1, 0) is also an eigenvector, but with eigenvalue 2. *Point 1.*
2. **C.** $A = U \Sigma V^T$: turn, stretch, turn. *Point 2.*
3. **B.** Without centring, the first direction just points at the middle of the cloud. *Point 3.*
4. **B.** 10 × (100 + 200 + 1) = 3,010 of 20,000 numbers: about 15%. *Point 4.*
5. **C.** $\log_2 8 = 3$, because 2 × 2 × 2 = 8. *Point 5.*
6. **B.** A fair coin is the hardest to predict: 1 bit. *Point 5.*
7. **B.** KL = cross-entropy − entropy = 2.0 − 1.5. *Point 6.*
8. **B.** ln 2 = 0.693 is the loss of saying 50/50 every time. *Point 7.*
9. **D.** $1 - 0.9^{100} \approx 99.997\%$. *Point 9.*
10. **B.** Real digits fill only a thin part of the 64-dimensional space; noise fills all of it. *Point 10.*